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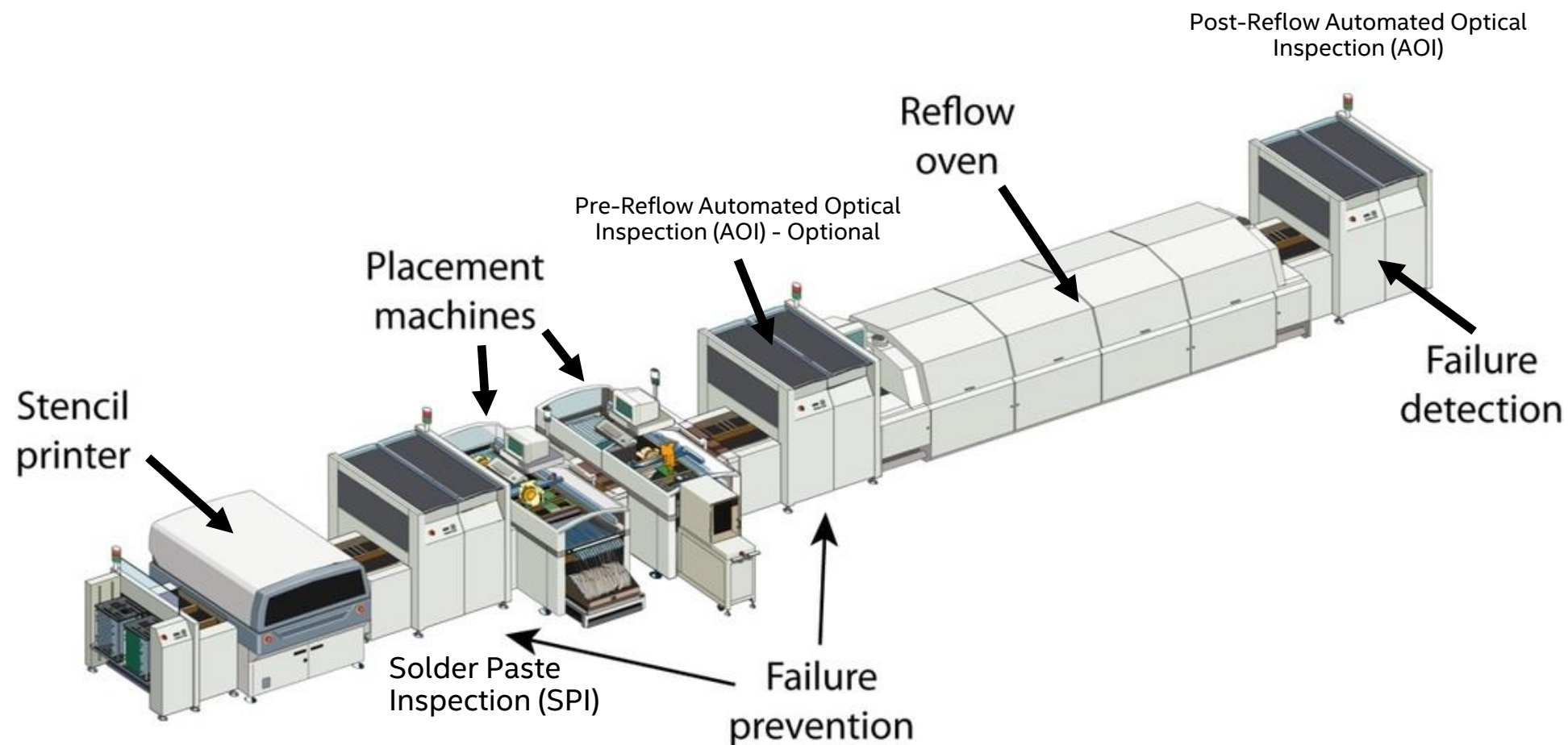
2023 WIFI Module Customers CRD Summary Report

Rev2.0, CQ&R CMEE Team, 2023Q2



intel®

General SMT Line Layout



Objectives

Share Findings From CMEE Mobile Industry Data Collection (WIFI Module Customer Survey)

- Data Collected From 10 ODM Customers/Sites
 - Compal (KunSan – Build for Dell/HP, ChengDu – Build for Dell, ChongQing – Build for Acer)
 - HuaQin (Build for HuaWei/Asus/Lenovo/Samsung)
 - Inventec ChongQing (Build for HP/Asus)
 - LCFC (Build for Lenovo)
 - MSI KunSan (Own Brand)
 - Pegatron (Suzhou – Build for Microsoft, ChongQing – Build for Asus)
 - Quanta (ChongQing – Build for HP)

Highlights

- Most of the customers follow Intel MAS to design land pattern and stencil.
- 4 mil is most popular stencil thickness, half of the customers use step-stencil on mother board WLAN SD1216 location.
- Most of the customers locate the module in corner or edge. No preference on top or bottom
- Only Compal CQ and HuaQin scan the SN# and save to database, and only Compal CQ's scanner has ECC feature.

WIFI Module (1 Of 4)

Are you following recommended WLAN SD1216 land pattern in MAS? If not, why?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	Yes	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
No, used to have solder open and short failures	Yes	Yes	Yes	Yes
Are you following recommended stencil opening for WLAN SD1216 in MAS?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	Yes	Yes	Yes	Yes
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
No, used to have solder open and short failures	Yes	Yes	Yes	Yes

Most of the customers follow Intel MAS to design land pattern and stencil.

WIFI Module (2 Of 4)

What is the preferred stencil thickness for WLAN SD1216?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
4 mil	5 mil	4 mil	4 mil	4 mil
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
4 mil	4 mil	4 mil	4 mil	4 mil
Is there a stencil thickness stepping up or down on mother board WLAN SD1216 location?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Yes	Yes	Yes	No	No
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
No	No	Yes	Yes	Yes

Most of the customers use 4 mil stencil, half of the customers use step-stencil on mother board WLAN SD1216 location.

WIFI Module (3 Of 4)

Where is the WLAN SD1216 located in your board design?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Edge	Edge, center	Edge	Edge	Edge
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
Edge	Corner	Edge	Edge	Edge or corner
Do you place the WLAN SD1216 on top (1 time reflow) or bottom (2 times reflow) side of the motherboard?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Both	Top	Bottom	Bottom	Bottom
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
Bottom	Bottom	Bottom	Top	Both

Most of the customers locate the module in corner or edge. No preference on top or bottom

WIFI Module (4 Of 4)

Do you scan the SN# of Intel SDWLAN module, and link it with board SN# and save it into your database?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
No	Yes	No	Yes	No
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
No	No	No	No	No
If yes, is your scanner having Error correction code (ECC) feature?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	Yes	NA	No	NA
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
NA	NA	NA	NA	NA

Only Compal CQ and HuaQin scan the SN# and save to database, and only Compal CQ's scanner has ECC feature.

Solder Paste Related(1 Of 2)

Can you provide your current SAC solder paste list with percent and usage?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
YunNan Tin: YW9-3005-98CY04,100%	Shenmao PF606-P 100%	Shenmao PF606-P, 50%/ YunNan TinYW9-3005-98CY04, 50%	ShenMao PF606-P, 73%;Eunow EUP-100F,27%	ShenMao PF606-P, 50% tamura LTF-204-931VT,50%
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
Shenmao 95%	KOKI S3X58-M406-3,100%	YunNan Tin YW9-3005-98US04 100%	Shenmao PF606-P,100%	Senju M705-S101HF(N4)-S4, 50% ,Eunow EUP-1000HQR: 30%
Can you provide your current/future LTS solder paste list with percent and usage?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	Nihon B37 100%	NA	No HVM plan, verified Eunow DW1000 and YunNan Tin YT-AB02
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
Senju and Shenmao 90% total	NA	NA	NA	NA

Different ODMs have different SAC solder paste preference.
Most ODMs don't have LTS experience

Solder Paste Related(2 Of 2)

Do you use LTS for other product except BGA? What paste do you use? What kind of product?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
NA	NA	No	NA	NA
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
Full board LTS	NA	NA	NA	NA
What would be the tolerance limits used on the SPI machine for 0.35 mm pitch part?				
Compal CD	Compal CQ	Compal KS	HuaQin	Inventec CQ
Volume:50%~180%	Volume:50%-150%	Volume:60%-160%	Volume:60%-150%	Volume:60%-140%
LCFC	MSI	Pega CQ	Quanta CQ	Pega SZ
Volume:70%-170%	No answer	Volume:50%-180%	Volume:50%-150%	Volume&Area:50%-150% Height: +2mil, -1mil

Only LCFC and Compal KS have LTS HVM experience

Improvement Suggestions from ODMs

Pegatron SZ:

- Change inner row pads from SMD to MD, to reduce SBB risk
- 2 connectors were easy to broken



Huaqin

- Reduce the distance between the pad(outer row) and edge to facilitate the reflow

